AcessGuru

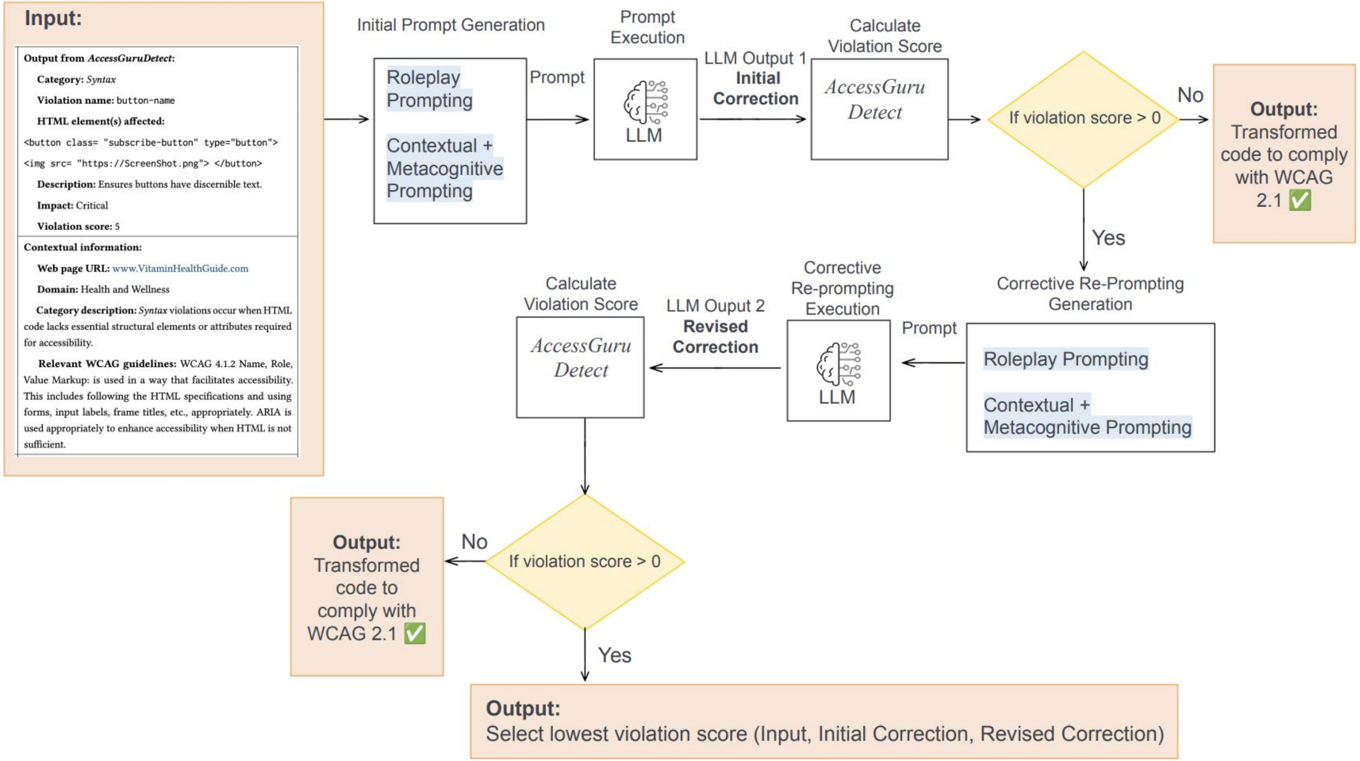
Problem

- Existing LLM-based accessibility detection tools lac *intent awareness*
  - Basically, they can identify a violation, but won’t process the actual purpose of elements
- Single-agent LLMs struggle with:
  - Complex structural corrections
  - Context-sensitive accessibility fixes
  - Maintaining DOM consistency
- No structured multi-agent reasoning framework existed for accessibility correction.

What they did

- Developed Multi-Agent framework:
    - Intent agent: Understands the developer’s or page’s intent.
    - Analysis Agent: Examines the violation and surrounding DOM structure.
    - Correction Agent: Generates accessibility-compliant HTML fixes.
    - Validation Agent: Verifies corrections and checks for new violations.

Figure 3: Overview of *AccessGuruCorrect* in: For each Web accessibility violation detected by *AccessGuruDetect* (examples of the input accessibility violations are shown in Tables 1 and 2), the LLM is prompted to generate the corrected code. The generated code is assigned a violation score; if the violation score remains above zero, corrective re-prompting is applied to improve the response further.



Research Gap

- Previous systems made corrections without actually accounting for the semantic purpose, developer’s intent, or functional role of elements.
- Previous research worked well on Alt-text detection but not structural/visual violations.
- Previous systems didn’t have a verification system to maintain correctness across multiple changes.

Evaluation

- Compared single agent systems to multi-agent systems
- Multi-agent systems far outperformed single.
    - Higher correction success for:
        - Structural violations
        - Landmark errors
        - Role conflicts
    - Better overall reasoning and understanding of the DOM hierarchy
    - Introduced fewer issues with suggestions.

Future Scope

- Improving communication between agents
- Handle layout/visual issues better
- Increasing scale and turning into an actual public-accessible tool.